

PHARMACIE TRAMED

Digital Transformation & AI Implementation Project

Computer Science Internship — Project Brief

April 2026

This document is a working project brief for the Pharmacie Tramed digital transformation initiative. It is still in active brainstorming phase and will be refined as the project progresses.

1. Business Information

Business Name	Pharmacie Tramed
Address	KN 151 St, Kigali, Rwanda
Phone	0788 301 960
Operating Hours	Opens 8:00 AM daily
Current Software	Ishyiga POS — installed, partially utilized
Technical Supervision	Systems Architect Engineer — Kigali, Rwanda
Strategic Advisory	Business Consultant — California, USA
Intern Role	Computer Science student — AI & Software Implementation

2. What Ishyiga Already Does

Ishyiga is developed by Algorithm Inc. and is the largest pharmacy management software provider in Rwanda and Burundi. It already includes several built-in capabilities, some of which are currently underused at Tramed. The table below shows what Ishyiga covers and where custom AI and software development adds value on top.

Ishyiga Feature	Available	⚡ What We Build On Top
Retail stock management & tracking	Yes	AI demand forecasting layer on top of stock data
Point of sale (POS) & sales recording	Yes	Sales pattern analysis —, top products, trends
Automated stock refilling alerts	Yes	Enhance with predictive reorder based on sales velocity
Cloud access via data centre (Ishyiga Live)	Yes	API integration for custom AI tools on top
Customer records & relationship tracking	Partial	Full CRM — chronic medication reminders, loyalty system

WhatsApp / customer communication	No	WhatsApp Business API automation we build
Delivery management	No	Delivery tracking module we build
Custom reporting dashboard	No	Daily summary dashboard we build on top of Ishyiga

Key insight: Ishyiga handles the core transactions. The role of this internship is to build the intelligence layer on top — turning raw Ishyiga data into predictions, automations, and insights that increase profitability.

3. The Gaps We Are Closing

While Ishyiga provides a solid foundation, several critical gaps exist that require custom software and AI solutions:

3.1 Data exists but no system extracts insights from it

Ishyiga records every sale, but there is currently no system to transform that data into meaningful insights. There is no dashboard, no trend analysis, and no forecasting — the data sits unused.

3.2 Stock management is experience-based, not system-driven

The pharmacy has operated for years and the experience built up over that time is genuinely valuable — the team has a strong feel for what sells and when. However, there is currently no digital system that captures and formalizes that knowledge. Ishyiga has refilling alerts but there is no predictive layer that uses historical data to anticipate demand before stock runs low. The goal is not to replace experience but to give it a system that makes it scalable and reliable.

3.3 No customer retention system

There is no WhatsApp automation, no chronic medication reminder system, and no loyalty program. Patients on recurring prescriptions have no structured follow-up, which means retention depends entirely on the customer remembering to return.

3.4 No AI layer

Ishyiga is a solid transactional system but has no predictive or intelligent features. There is no demand forecasting, no sales pattern recognition, and no anomaly detection. This gap is exactly what a Computer Science internship is positioned to fill.

4. What We Build — CS Internship Scope

This section defines what a student builds during this internship, divided by technical complexity from no-code to full AI engineering. The tools list is subject to revision based on technical guidance received during the project.

4.1 No-Code Implementations (Independent)

- Ishyiga full activation — ensure all available features are set up and used consistently
- WhatsApp Business setup — automated replies for stock availability queries
- dashboard — daily sales summary drawn from Ishyiga data
- Make / Zapier automations — connect Ishyiga alerts to WhatsApp and supplier messages
- Customer loyalty tracker — simple digital system for repeat customer rewards

4.2 Software Engineering (Guided by Technical Supervisor)

- Data extraction pipeline — pull sales data from Ishyiga, clean and structure it for analysis
- Demand forecasting model — predict stock needs using historical sales patterns
- Custom reporting dashboard — web-based daily summary showing revenue, margins, alerts, trends
- Chronic medication reminder system — automated WhatsApp reminders tied to purchase history
- Stock anomaly detection — flag unusual discrepancies automatically
- Supplier automation scripts — auto-generate reorder messages when stock hits minimum

4.3 AI / Machine Learning Layer

- Sales trend analysis — identify which products sell more during certain seasons or events
- Customer segmentation — group customers by purchase behaviour for targeted communication
- Profit margin optimization — recommend which products to prioritize based on margin data
- Predictive restocking — forecast exact quantities to order per supplier per cycle

5. Internship Structure

The internship is structured in three phases over approximately 4 months:

Phase	Title	Deliverables
Phase 1	Audit & Discovery	Technical discovery audit of Ishyiga usage, operations, stock, and customer flows. Produce written diagnosis report and priority list.
Phase 2	Implementation	Build no-code solutions independently. Develop software and AI components under technical supervision. Show progress every 2 weeks via video demo.

Phase 3	Results & Packaging	Measure what improved. Write final case study. Package solution as replicable product for other pharmacies. Receive formal internship documentation.
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6. Supervision & Agreement

Intern	Computer Science student — AI & software implementation
Supervisor Deliverable	Technical guidance, coding support
Intern Deliverable	Audit report, working software, AI models, final case study

7. Tools & Technology Stack

Tools are chosen for their relevance to the Rwandan market and their employability value for a CS student. The final stack will be confirmed with the technical supervisor at the start of the internship.

Tool	Purpose
Ishyiga POS	Core pharmacy management platform
Backend technology	To be discussed with the supervisor
Front end technology	To be discussed with the supervisor
VSCode + Claude code	Development environment + AI coding assistant
WhatsApp Business API	Customer communication automation
Make / Zapier	No-code workflow automation
Google Sheets / Data Studio	Dashboards and reporting
Git / GitHub	Version control for all code

8. Bigger Picture — From Internship to Startup

8.1 Pharmacie Tramed as Pilot

Tramed is the first client and the testing ground. Every solution built here is documented as a replicable blueprint. The goal is not just to improve one pharmacy — it is to build a product that works in any pharmacy.

8.2 Replication & Commercialization

Once the pilot is proven at Tramed, the same system is packaged and offered to other pharmacies in Kigali and beyond. The case study from Tramed becomes the proof-of-concept document for every future client. This is the foundation of a healthcare AI startup in Rwanda — a commercial product born directly out of this internship.

8.3 Scaling Strategy

Once the product is proven at Tramed, business development and client acquisition expertise becomes the key asset for growth. The project is designed to be approached commercially — how to price the consultancy service, how to structure the offer for replication, and how to approach new pharmacy clients across Rwanda.

9. Why This Matters

This internship is not a generic placement. It produces real, measurable outcomes directly relevant to the technology job market:

- Real client, real data, real impact — not a simulated academic project
- Full software development cycle — from discovery and audit through to deployment and documentation
- AI and machine learning applied in a live business environment
- API integration, automation engineering, and data pipeline development
- A documented case study and GitHub portfolio that any employer can verify
- Supervised by a qualified Systems Architect Engineer
- Potential to scale into a funded healthcare AI startup — real entrepreneurial exposure

The skills developed here map directly onto roles in digital transformation, health technology, data engineering, AI implementation, and software consultancy — all high-demand areas in Rwanda and internationally.

This document is a living brief. It will be updated as the audit is completed, the technical supervisor provides direction, and the project scope is finalized. All sections marked as brainstorming will be confirmed before implementation begins.
